

Oswin Franco Cervantes

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Statement of Purpose

My goal is to use first-principles physics, electromagnetic modeling, circuit and system-level design, measurement, and computational methods to develop integrated hardware spanning semiconductor circuits, RF, photonics, and quantum systems.

Education

Massachusetts Institute of Technology, Cambridge, MA

Expected May 2028

B.S. Candidate in Physics and Electrical Engineering with Computing

Sept. 2024–Present

Selected coursework: **Graduate:** High-Frequency Integrated Circuits, Electromagnetics; **Coursera:** Microwave Engineering and Antennas, RF and Millimeter-Wave Circuit Design; **EE/Physics:** Semiconductor Electronic Circuits, Signal Processing, Quantum Physics I, Statistical Mechanics; **Computer Science:** Mathematics for Computer Science, Computational Structures, Low-Level Programming

Georgia Tech, Atlanta, GA — Dual Enrollment, Mathematics/Computer Science; GPA: 4.0/4.0

Aug. 2022–May 2024

Integrated Circuit Design Project

22 nm FinFET Amplifier and Control Chip

Spring 2026

MIT Semiconductor Electronic Circuits

Cambridge, MA

- Designed and simulated a 22 nm FinFET CMOS chip combining an analog amplifier, activation switches, and digital control logic; applied semiconductor device physics and large- and small-signal models.
- Created the chip layout in Cadence for a course tapeout and ran Calibre DRC/LVS checks to verify fabrication rules and consistency with the schematic.
- Verified through simulation that the amplifier exceeded targets of 26 dB gain at 1 MHz and 300 mV peak-to-peak output swing while operating 48% below its 500 μ W power limit.

Experience

Undergraduate Researcher — Prof. William D. Oliver

Sept. 2025–Present

MIT Research Laboratory of Electronics, Engineering Quantum Systems (EQuS) Group

Cambridge, MA

- Serve as technical owner for the on-chip microwave-filter subsystem within a cross-functional quantum-hardware team.
- Design filters for operation from room temperature to cryogenic conditions, applying transmission line and impedance matching principles to suppress out-of-band noise while minimizing passband insertion loss at the interface between semiconducting quantum-dot qubits and superconducting circuits.
- Use PyAEDT to automate 3D model generation and HFSS Driven Terminal and Eigenmode simulations for S-parameter and resonant-mode analysis; use Q3D for quasi-static parasitic extraction and AWR Microwave Office for circuit models.
- Maintain the filter component library in CDES, an object-oriented Python framework; use NumPy, pandas, and gdstk to automate generation of fabrication-ready GDS layouts.
- Fabricate superconducting niobium microwave filters on silicon substrates using photolithography in a cleanroom environment.

Avionics Engineer

Jan. 2026–Present

MIT Rocket Team

Cambridge, MA

- Design RF circuits and PCB interconnects in Altium; analyze interconnect behavior with HFSS Driven Terminal and Ansys SIwave; compare simulated S-parameters with vector network analyzer measurements.

Hardware Development Engineer (Systems Engineering) Intern

May 2025–Aug. 2025

Amazon Lab126, Advanced Products

Sunnyvale, CA

- Collaborated with a cross-functional team to translate product requirements into system architecture and functional block diagrams, then designed schematics and PCB layout for a multi-IC solar-charging module for Kindle devices.
- Designed and validated buck, boost, and buck-boost power circuits using OrCAD and SPICE for circuit design and simulation and Cadence Allegro for PCB layout; characterized prototypes with oscilloscopes and electronic loads and investigated maximum power point tracking (MPPT).
- Implemented I²C communication and assembled SMD prototypes to verify end-to-end circuit functionality; trained new hires on PCB design tools.

Technical Skills

Design and Analysis: Electromagnetic modeling, RF/microwave circuits, analog/mixed-signal ICs, power circuits, S-parameters, transmission lines, impedance matching, SPICE, PCB/system design

Fabrication and Test: 22 nm FinFET custom layout, DRC/LVS, GDS/KLayout, niobium-on-silicon photolithography, cleanroom processing, SMD, VNA and oscilloscope measurements

Software and HDL: Python, C, Java, Julia, MATLAB, Bash, VHDL, Bluespec HDL, RISC-V assembly; NumPy, pandas, gdstk, Git

EDA Tools: Ansys (HFSS, Maxwell, Q3D, SIwave, PyAEDT); Cadence (custom IC, AWR Microwave Office, Allegro); Calibre, Altium, OrCAD, KLayout

Honors and Certification

Honors: QuestBridge Scholarship; Jack Kent Cooke Foundation Semifinalist; Stanford Intrologic Scholar

Certification: TestOut Network Pro Certification